

# **Basic Principles of Medicine 1**

## **Module: Cardiovascular, Pulmonary and Renal**

### **Lecture: CPR 09**

### **Hemodynamics of Pressure, Flow and Resistance**

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# Recommended readings

Chapter 13: Mechanics of the Cardiac Pump. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell.

Klabunde Cardiovascular physiology concepts. 2<sup>nd</sup> ed. Lippincott. Chapter 5: pp 93-108

Review: DLA and Lecture: Introduction to the cardiovascular system

- This lecture discusses the physical laws that govern the flow of blood in the blood vessels.

*Slides with blue background are the supplemental information associated with the previous slides and thus, might not be explained during the lecture time. However, please read the slides for clarification as they are equally important to know.*

# Learning Objectives

**SOM.MK.II.BPM1.3.CPR.2.PHYS.0108** Contrast flow and velocity and list the measurement units for each.

**SOM.MK.II.BPM1.3.CPR.2.PHYS.0109** Explain how Darcy's law maps the relationship between pressure, flow, and resistance in the vasculature and be able to calculate for one variable if the other two are known. Apply this relationship to the arteries, arterioles, capillaries, venules, and veins. Explain how blood flow to any organ is altered by changes in resistance to that organ.

**SOM.MK.II.BPM1.3.CPR.2.PHYS.0110** Discuss how Poiseuille's Law explains the changes in resistance with flow. Use it to calculate changes in resistance in a rigid tube (blood vessel). Explain the deviations from Poiseuille's law predictions that occur in distensible blood vessels.

**SOM.MK.II.BPM1.3.CPR.2.PHYS.0111** Discuss the relationship between flow, velocity and cross-sectional area. Determine the changes in these variables that occur with changes in vascular compliance.

**SOM.MK.II.BPM1.3.CPR.2.PHYS.345** Contrast resistance and conductance in the vascular system. Understand the effects of adding resistance in series vs. in parallel on total resistance and flow.

**SOM.MK.II.BPM1.3.CPR.2.PHYS.0112** Apply this information to solving problems characterized by a) resistances in series and b) resistances in parallel. Apply this concept to the redistribution of flow from the aorta to the tissues during exercise.

# Lecture Outline

## **Darcy's Law/ Bulk Flow law ( $FLOW = \Delta P / R$ )**

### Poiseuille's Law

- Poiseuille's law which states that flow (Q) is equal to pressure (P) [the change in pressure, or the difference between mean arterial pressure (MAP) and central venous pressure (CVP)] divided by resistance (R)

### Importance of resistance

- Resistance arterioles controls flow of blood to organs
- Parallel vs series

### Velocity, Flow and cross-sectional area

- As the blood vessels cross sectional area increases, flow increases, the velocity decreases.

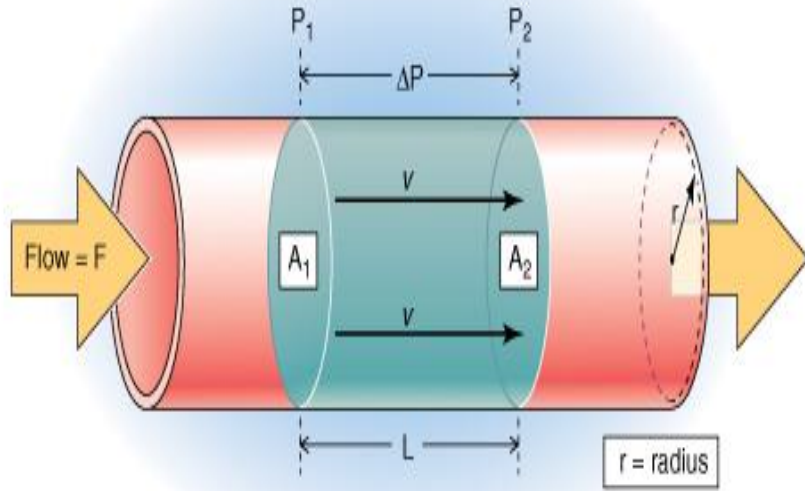
# Points to consider

Hemodynamics is the study of: **Blood Flow, Pressure, Resistance**

Consider: the rules of the cardiovascular system

1. Why does blood flow round the circulation? **Bulk flow from High to low pressure states**
2. What factors affect flow? Volume of blood that passes a given point per unit time(L/min). **Determined by pressure gradients and resistance.**
  - Resistance opposes blood flow. Depends on length of the tube, Radius of the tube ( $4^{\text{th}}$  power) and Viscosity of blood
3. What determines velocity of blood flow? How far a volume of blood travels per unit of time (mm/sec). **Determined by cross-sectional area, if the flow rate is constant.**

# 1. Bulk Flow Law or Darcy's Law



1. FLOW  $\propto$   $P_1 - P_2$  **or**

i.  $(\Delta P) = k \cdot (\Delta P)$

$k$  = hydraulic conductance

2. FLOW  $\propto$   $1/\text{Resistance}$

$$\text{FLOW} = \Delta P / R$$

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**Pressure ( $\Delta P$ ):** driving force to move fluid through a tube of constant diameter/radius and known length

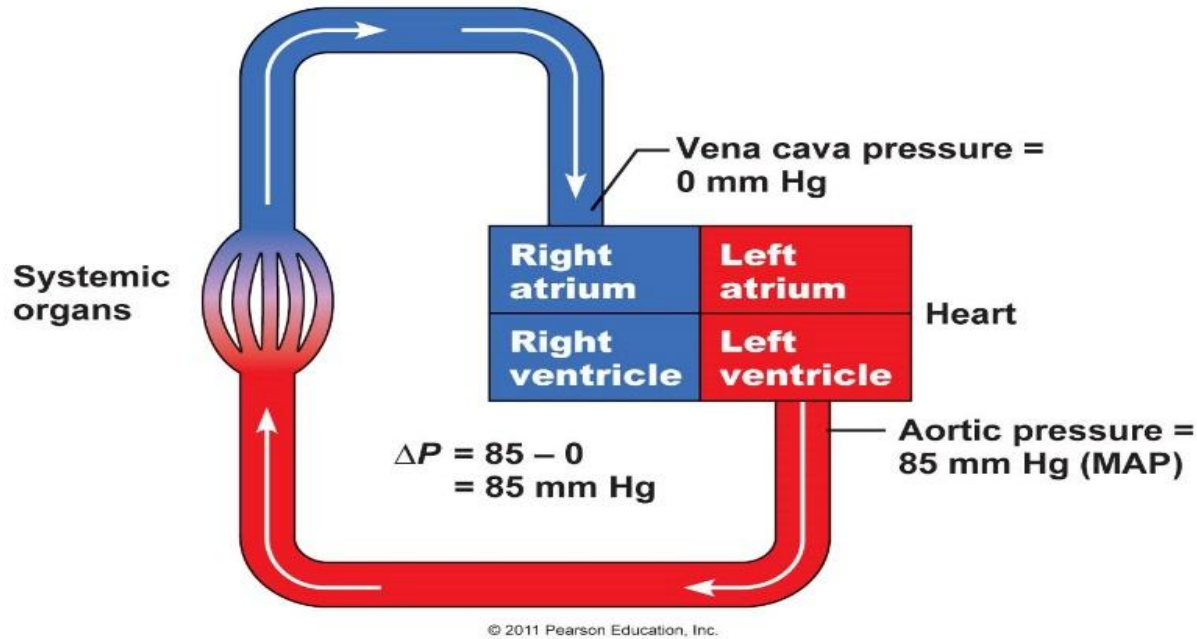
**Resistance ( $R$ ):** used to regulate flow in body

**Flow ( $Q$ ):** volume/time

## \* Clarification: types of “pressure” in CV Physiology

- **Driving pressure**
  - Pressure difference along the length of a vessel
  - Responsible for blood flow
- **Transmural pressure**
  - Pressure difference across a blood vessel wall
  - Influences vessel diameter because vessels are compliant
- **Hydrostatic pressure (important in capillaries)**
  - Pressure created by the weight of blood
  - Increases when moving from supine to standing

# Pressure Gradient ( $\Delta P$ ) and Resistance (TPR)



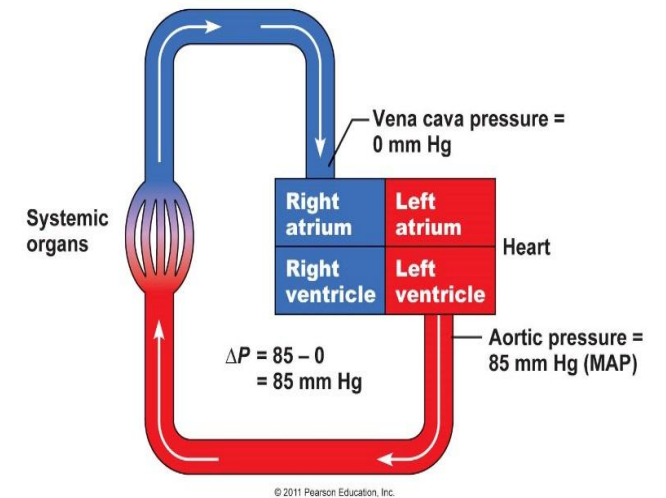
- Blood Flows around systemic circulation because of the pressure gradient between the Aorta and Right Atrium
- Pressure difference between an upstream and downstream points due to resistance between these points.

- Driving Pressure/pressure gradient
  - $\Delta P = \text{Driving pressure} = \text{Mean arterial pressure} - \text{Right atrial pressure}$ 
    - $\Delta P (\text{Systemic}) = 85 - 0 (85)$
    - $\Delta P (\text{Pulmonary}) = 25 - 8 (17)$
- Systemic Blood flow = Pulmonary blood flow = 5L/min ... Why?

- Resistance
- $\Delta P = \text{Flow} \times \text{Resistance}$
- $\Delta P = \text{Flow} \times \text{Resistance}$ 
  - Systemic vascular resistance = 0.02 mmHg x min/ml
  - Pulmonary vascular resistance = 0.003 mmHg x min/ml

# Pressure Gradient ( $\Delta P$ ) and Resistance (TPR)

- Calculating Flow, Driving Pressure or resistance
- **Flow** through the systemic and pulmonary circulations = Cardiac output (Q)
- 1.  $\Delta P$  (systemic circulation) = MAP- RAP (CVP)
  - Assuming RAP =  $\sim 0$ ; then  $\Delta P$  = MAP
- 2. R (systemic circulation) = Total peripheral resistance (TPR)/ systemic vascular resistance (SVR)
- **Flow**=  $\Delta P / R$  becomes:  **$CO(Q) = MAP - RAP / TPR$**  Variables are interchangeable
- Easier to measure pressure than flow (vol/time) in body.
  - **MAP = CO x TPR**
  - **TPR = MAP-CVP /CO** (\*CVP $\sim 0$ )
- MAP is influenced by both the heart (via CO) and the vasculature (via TPR)
- If CO increases then MAP increases if TPR is constant
- If TPR increases then MAP increases if CO is constant
- However, TPR is not determined by MAP and CO
- What determines the Resistance???



- **Practice question:**
- If MAP is 100 mmHg, and CVP is zero, and CO is 6 L/min –what is the value of the TPR?
- **Ans:**  $100/6 = 16.7 \text{ mmHg} \times \text{min/ml}$

# Lecture Outline

Darcy's Law/ Bulk Flow law (**FLOW =  $\Delta P / R$** )

## Poiseuille's Law

- **Poiseuille's law which states that flow (Q) is equal to pressure (P) [the change in pressure, or the difference between mean arterial pressure (MAP) and central venous pressure (CVP)] divided by resistance (R)**

Importance of resistance

- Resistance arterioles controls flow of blood to organs
- Parallel vs series

Velocity, Flow and cross-sectional area

- As the blood vessels cross sectional area increases, flow increases, the velocity decreases.

## 2. Poiseuille's Law

Relates 1. resistance to fluid flow in a tube to  
2. geometry of tube and  
3. properties of fluid.

**Darcy's Law:** Flow =  $\Delta P / R$

**Poiseuille defined: Flow** =  $\Delta P \times \frac{\pi r^4}{8\eta L}$  *8, length and  $\pi$  are constant for the circulation*

**Poiseuille defined:** Resistance  $\propto \frac{8 \times \text{viscosity} \times \text{length}}{\pi \times \text{radius}^4}$

**Poiseuille's Law** Resistance  $\propto \eta$  (viscosity)  
Resistance  $\propto 1/r^4$  (radius)

So, Flow  $\propto 1/\text{viscosity } (\eta) \text{ or length}$   
**Flow**  $\propto$  **radius<sup>4</sup>**

- Resistance is determined by vessel radius to the fourth power, length of vessels and viscosity of blood

## Summary: How does blood flow through systemic and pulmonary circulation?

Blood flows through the systemic circulation from the LV to the RA because the LV contracts and ejects the blood forcefully at a high pressure into the aorta. The **aortic blood pressure is ~ 120 mm Hg** (i.e. 120 mm Hg above atmospheric pressure). After passage of the blood through the systemic circulation, the blood returns to the **RA where the pressure is only ~ 3 mm Hg**. The contraction of the LV then creates a pressure gradient between the aorta and the RA which enables blood to flow down this gradient and through the circulation.

Similarly, the contraction of the RV forcefully ejects blood into the pulmonary artery. However, the pressure generated by the **RV is much less (~ 25 mm Hg)** than that generated by the LV. The pressure in the pulmonary vein and **LA is ~ 8 mm Hg**. The pressures generated by the RV do not have to be as high as those generated by the LV since the resistance to the flow of blood in the pulmonary circulation is much less compared with the systemic circulation. The systemic circulation is a high-pressure circuit and the pulmonary circulation is a low-pressure circuit.\*

Because the heart is a pump – it ejects blood intermittently and so the aortic (arterial) pressure is pulsatile. When the LV contracts (systole) the pressure in the aorta is 120 mm Hg, when the heart relaxes (diastole) the pressure in the aorta does not fall to zero but falls only to 80 mm Hg. In the pulmonary circulation the systolic blood pressure in the pulmonary artery is 25 mm Hg and the diastolic blood pressure is ~8 mm Hg. Conventionally these are written as 120/80 mmHg and 25/10 mmHg respectively. **When blood pressure is measured routinely, it is the systemic BP that is measured.**

Although arterial pressure is pulsatile with a peak and a trough, most aspects of hemodynamics can be understood by considering the **mean arterial pressure which is ~ 93 mm Hg**. *For reasons given later, the mean arterial pressure is not simply the arithmetic mean between systolic (120) and diastolic (80).*

In the CVS: the larger the pressure gradient, the bigger the flow of blood (when resistance is constant) and the greater the resistance the smaller the flow.

\*Both the systemic and the pulmonary blood vessels offer resistance to blood flow. Most of the resistance is due to the friction between the blood components and the walls of the vessels. The diameter of the blood vessel is very important in determining the resistance to flow: **the smaller the diameter – the higher the resistance**, and vice versa.

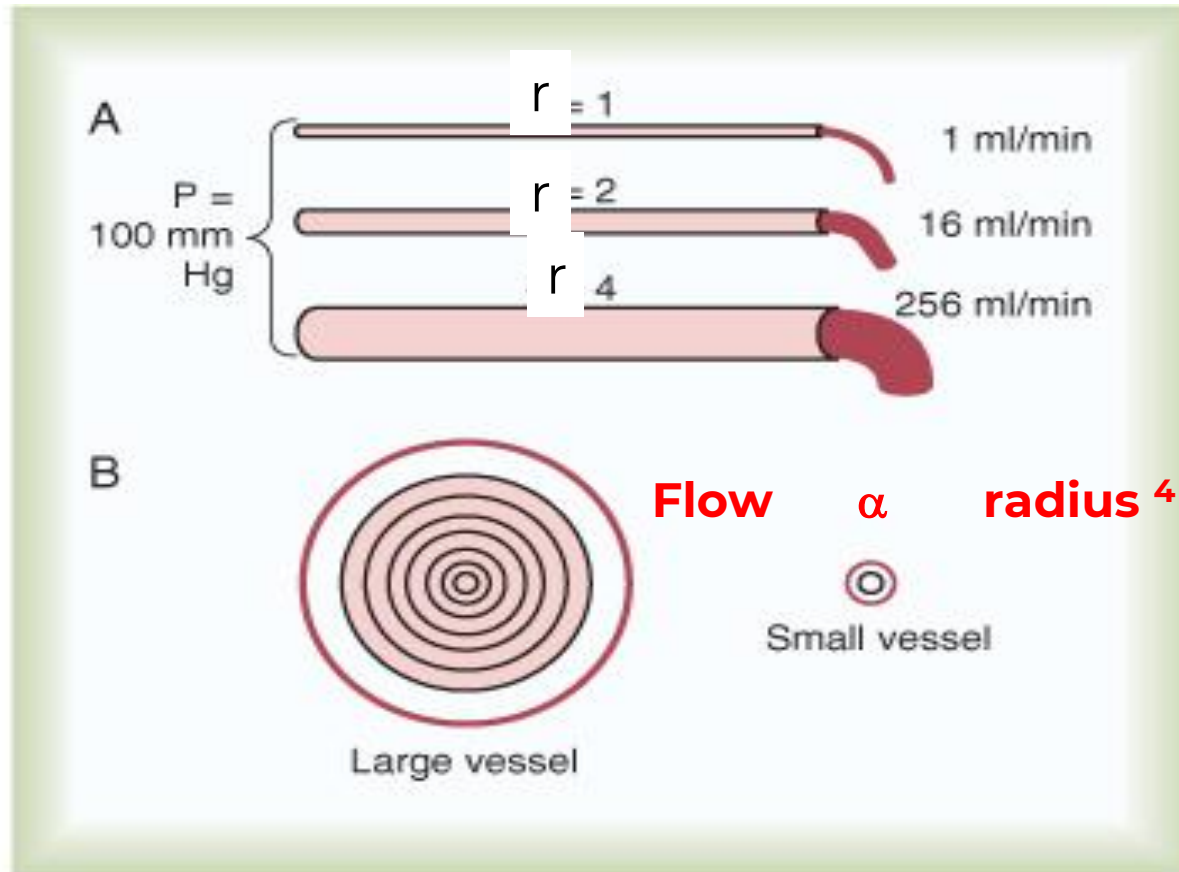
The total resistance of the systemic circulation is ~ 0.02 mmHg/ml/min; the pulmonary circulation is much lower – 0.003mmHg/ml/min. A low pulmonary artery pressure is usually sufficient to drive the cardiac output through the lungs whereas the LV has to generate a much higher pressure in the aorta to drive blood through the high resistant systemic circulation.

# Poiseuille's Law- resistance

**Resistance most sensitive to radius:  $1/r^4$**

(Decrease radius / constriction  $\rightarrow$  increase Resistance)

Small changes in vessel diameter results in large changes in flow of blood



# Deviations from Poiseuille's Law

Law can apply in general to blood flow in body.

Equation has following assumptions, and applies under the below conditions:

- 1. Incompressible:** Valid – blood is fluid which cannot be compressed.
  - 2. Laminar flow:** Valid: Blood flow is usually laminar – in vessels
- 
- 1. Rigid, cylindrical, unbranched tubes:** Vasculature is not.
  - 2. Non-pulsatile flow:** Blood flow pulsatile flow
  - 3. Newtonian fluid:** Blood is non-Newtonian. Hematocrit (% RBC volume in blood) alters viscosity and thus flow.

# Factors affecting resistance - Viscosity

- Resistance  $\propto$  **Viscosity ( $\eta$ )**
- *Viscosity = degree of slipperiness of 2 layers of fluid*
- Viscosity  $\propto$  Hematocrit
  - *Hematocrit = Packed Cell Volume/Total Blood Volume*
- When hematocrit is high, the resistance to blood flow is increased. (more work is expended driving blood round the circulation)

# Lecture Outline

Darcy's Law/ Bulk Flow law (**FLOW =  $\Delta P / R$** )

Poiseuille's Law

- Poiseuille's law which states that flow (Q) is equal to pressure (P) [the change in pressure, or the difference between mean arterial pressure (MAP) and central venous pressure (CVP)] divided by resistance (R)

## Importance of resistance

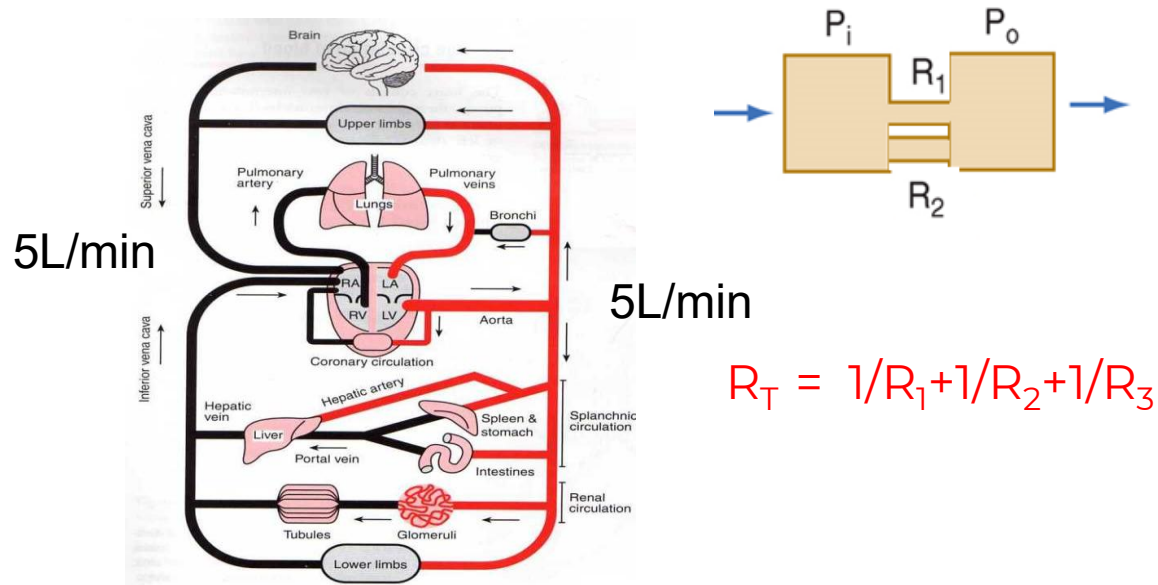
- **Resistance arterioles controls flow of blood to organs**
- **Parallel vs series**

Velocity, Flow and cross-sectional area

- As the blood vessels cross sectional area increases, flow increases, the velocity decreases.

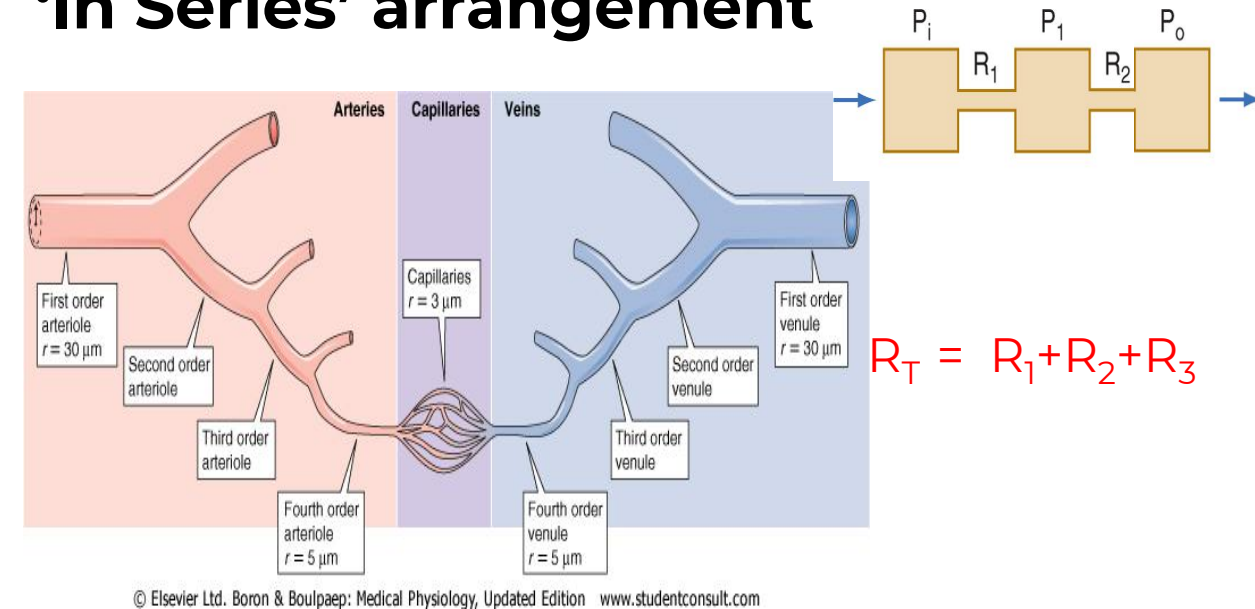
# Vasculature is arranged in two main ways to provide blood flow to organs

## 'In Parallel' arrangement



- Vascular networks of major distributing arteries of most individual organs are '**in parallel**' with other organ networks.
- Total resistance encountered by blood decreases. With multiple pathways available, the resistance to blood flow is reduced, leading to an increase in blood flow rate.

## 'In Series' arrangement

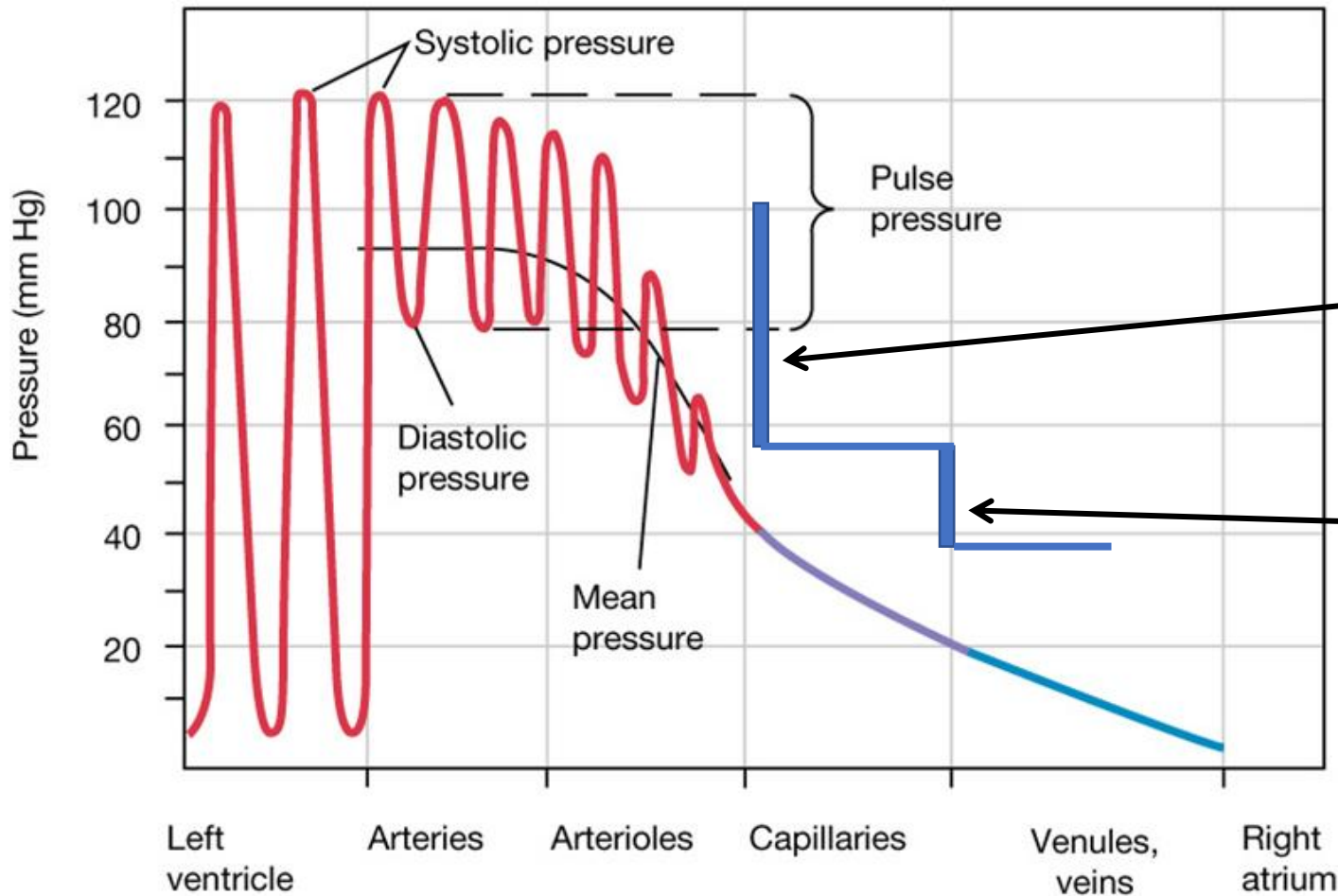


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- Within each organ, the blood vessels (arteries) divide into arterioles and into capillaries, then reunite into venules and veins.
- Within each organ, the resistances are '**in series**'
- however capillary beds/networks are also regarded as parallel networks

# Resistance arterioles: region of greatest pressure drop

Flow from: High pressure (aorta, arteries) → low pressure (capillaries) → Lowest pressure (right atrium)



Aorta: Large radius - resistance is low.  
Not much pressure difference needed for large flow.

Arterial pressure not ~ 0 during diastole, as in ventricle; recoil leads to diastolic pressure

Arteriole: Bigger Drop in pressure gradient ( $\Delta P$ ).  
50-70% drop. Resistance Greater!

Capillary: Smaller drop in pressure gradient ( $\Delta P$ ).  
Resistance Less!

Lowest pressures in venules and veins, capacitance vessels – unstressed blood

**WHY???**

Explanation of graph on previous slide:

From the diagram above you can see that there is very little pressure drop from the aorta to the large arteries; in fact from the aorta to the radial artery there is only a drop of  $\sim 2$  mm Hg. This is because these vessels have a large diameter, and so do not provide much resistance to flow.

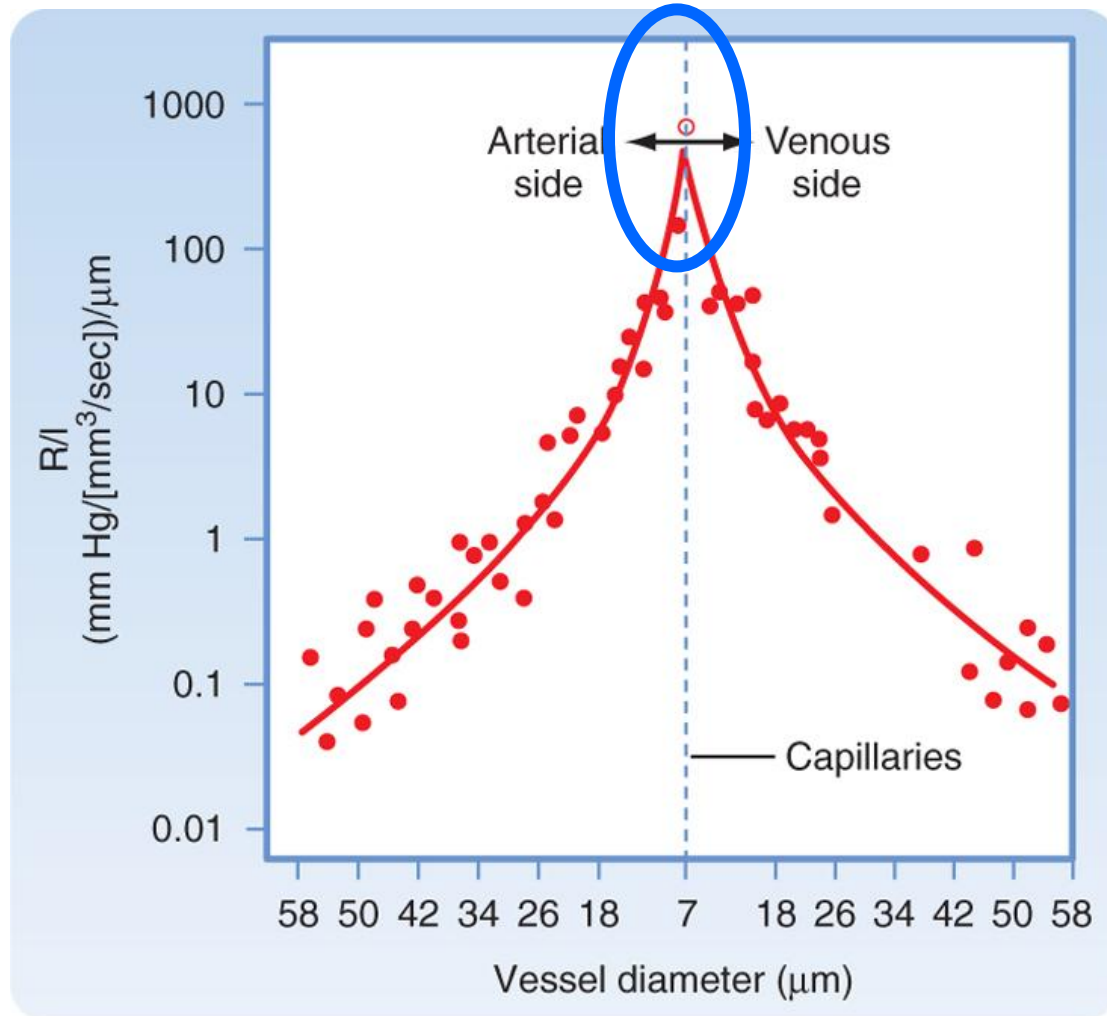
The largest fall in pressure is seen to occur across the smallest, terminal arteries and the arterioles. According to Darcy's law this must be because these vessels must offer the largest resistance to flow. This is the case, and this is why they are called resistance vessels.

The capillaries, although very narrow, surprisingly offer little resistance to flow and the pressure drop across them is minimal. This is due to the fact that there are very many of them and they are arranged mostly in parallel.

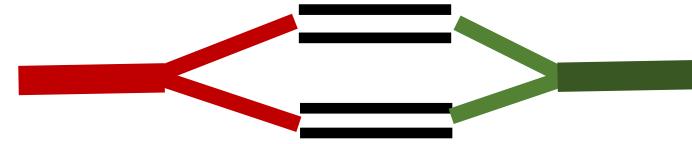
*(Remember your electricity, circuits with parallel resistances have a smaller total resistance than circuits with series resistances).*

Since there are many more venules and small veins compared with arterioles and small arteries their net resistance is low. The pressure in small veins is  $\sim 10 - 15$  mmHg and this is sufficient to drive the blood (cardiac output) from the veins to the right atrium.

# But.. Isn't resistance greatest in capillaries?!



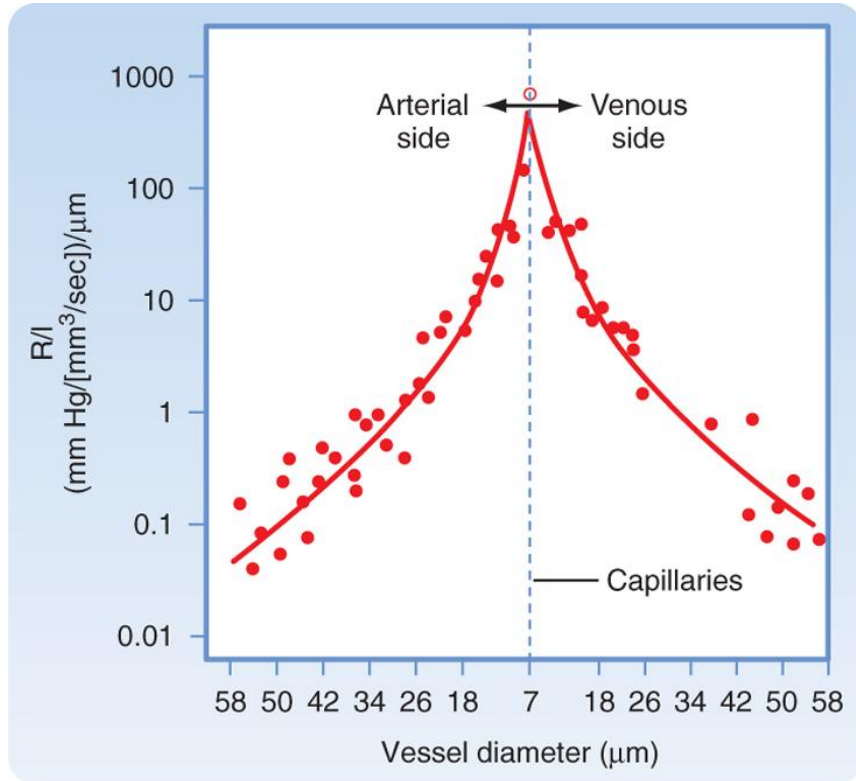
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Poiseuille's law: ( $R \propto 1/r^4$ )  
Capillaries have a smaller diameter than arterioles

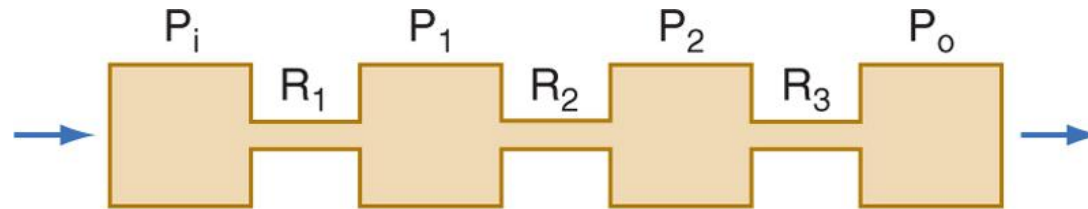
Resistance of ONE capillary >  
Resistance of ONE arteriole

# Yes.. But ..resistance is greatest in capillaries!



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## WHEN IN SERIES



Total Resistance ( $R_{total}$ )

$$= R_1 + R_2 + R_3 + R_4 + R_5$$

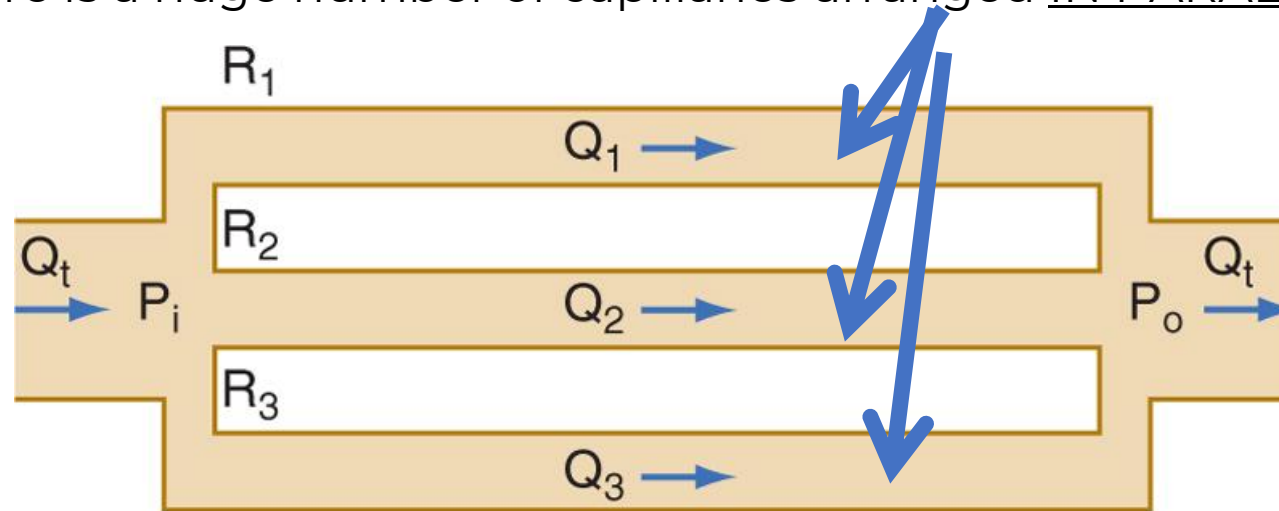
Total Resistance from  
artery to vein

=

Sum of the resistances of the  
individual components

# Why capillary network/bed NOT offer greater resistance to flow than arteriolar network?

Because there is a huge number of capillaries arranged IN PARALLEL (capillary bed).



The resistances are '**IN PARALLEL**'. Flow can occur through paths with least resistance. Flow in and out remains the same.

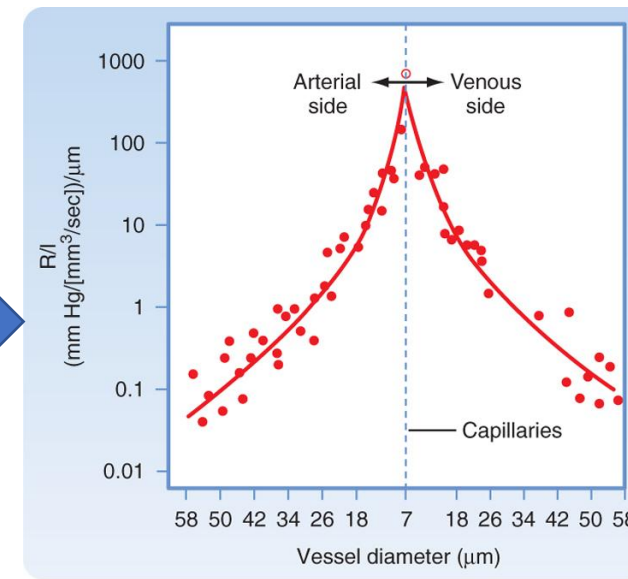
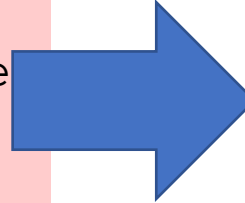
Total resistance across a parallel arrangement is less than the sum of the individual components.

$$C_{total} = 1/R_{total} \quad \text{Total conductance} = \text{sum of the individual conductances}$$

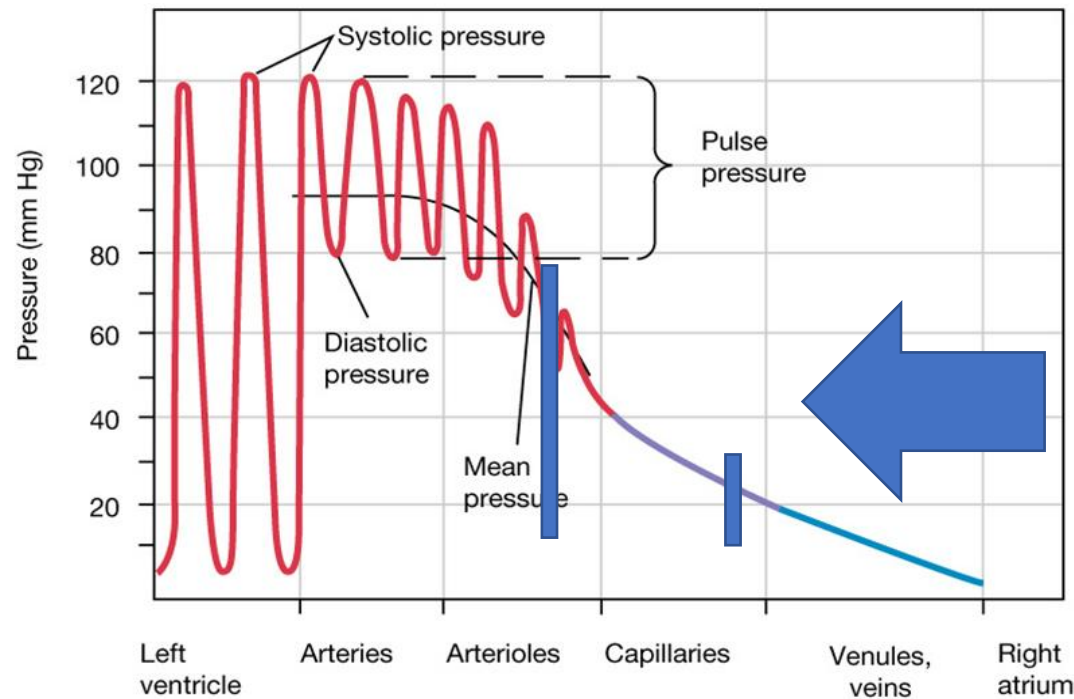
SO

Resistance in ONE capillary > ONE arteriole  
IN SERIES

although resistance of a single capillary is greater than the resistance of a single arteriole because the capillaries are arranged in parallel they as a group, offer less resistance to flow of blood.



Pappano U. Medical Cardiovasc. Physiol



Resistance in arterioles > capillaries (plural)  
due to PARALLEL arrangement of capillaries

Pressure drop greatest across  
resistance arterioles

Consequently, the pressure **drop** across arterioles is about 50 mmHg (i.e. from ~80 to ~30 mmHg) whereas the pressure **drop** across capillaries is about 18 mmHg (i.e. from ~30 mmHg to ~12 mmHg).

resistance changes by altering the diameter of the vessel

# Resistance: regulating flow to organs

|                  | Cardiac output at rest 5 L/min | Cardiac output during exercise 25 L/min |
|------------------|--------------------------------|-----------------------------------------|
| Brain            | 15 %<br>0.75 L/min             | 3-4%<br>0.75 L/min                      |
| Heart            | 5%<br>0.25 L/min               | 5%<br>1.25 L/min                        |
| GI               | 55%<br>2.75 L/min              | 10%<br>2.5 L/min                        |
| Kidneys          | 20%<br>1 L/min                 | 5%<br>1.25 L/min                        |
| Skeletal Muscles | 20%<br>1 L/min                 | 85%<br>21.25 L/min                      |

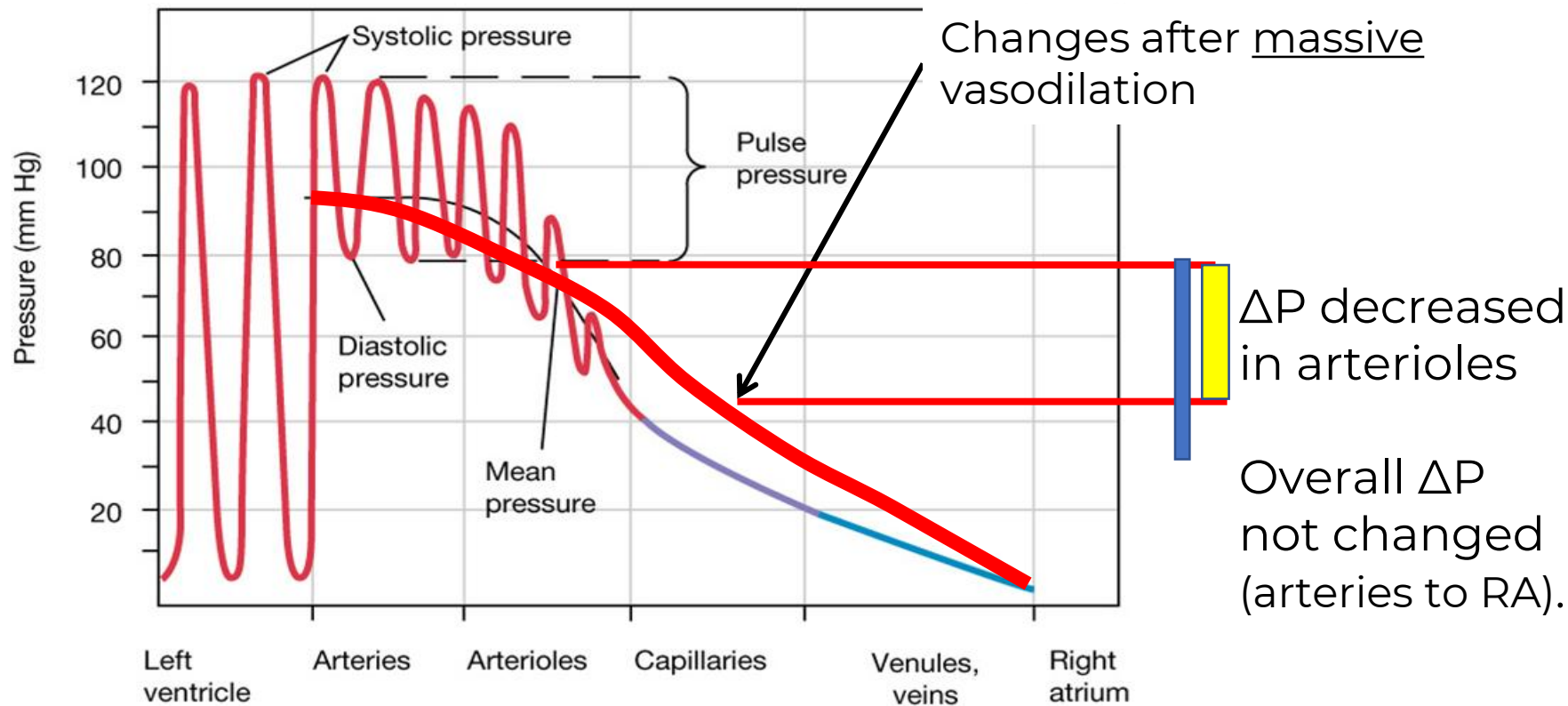
Flow through each organ is a fraction of the total blood flow.

Skeletal muscle receives 20%, exercise - increases to ~80%

Brain receives 13%,

Could blood flow to Skeletal muscle be altered?

# Resistance arterioles: Vasodilation



[http://faculty.pasadena.edu/dkwon/chapter%2015/chapter%2015\\_files/textmostly/slide12.html](http://faculty.pasadena.edu/dkwon/chapter%2015/chapter%2015_files/textmostly/slide12.html)

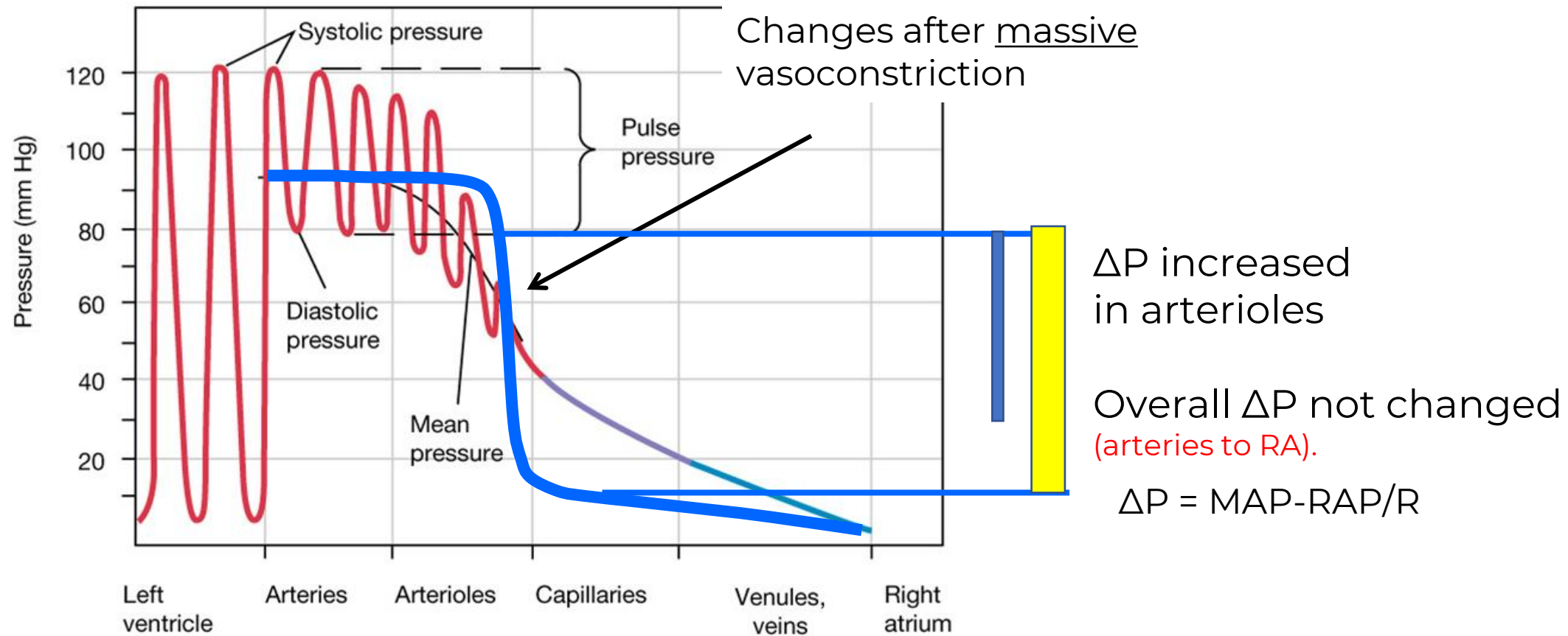
**Short term:** Elevated pressures at capillaries, increased edema.

**Clinical examples:** Septic shock, or anaphylaxis → massive vasodilation. However, (longer time than shown) there is eventually pooling of blood in capacitance vessels (venules, veins), hypotension, reduced tissue perfusion, changes in heart rate etc. [medical emergency].

**Exercise** - massive reduction of resistance- inc blood flow muscles. Redistribution of blood flow to other tissues to maintain MAP

Graves Disease/Hyperthyroidism: Vasodilation of resistance arterioles, Pulsatile flow in capillaries seen in fingernail beds

# Resistance arterioles: Vasoconstriction



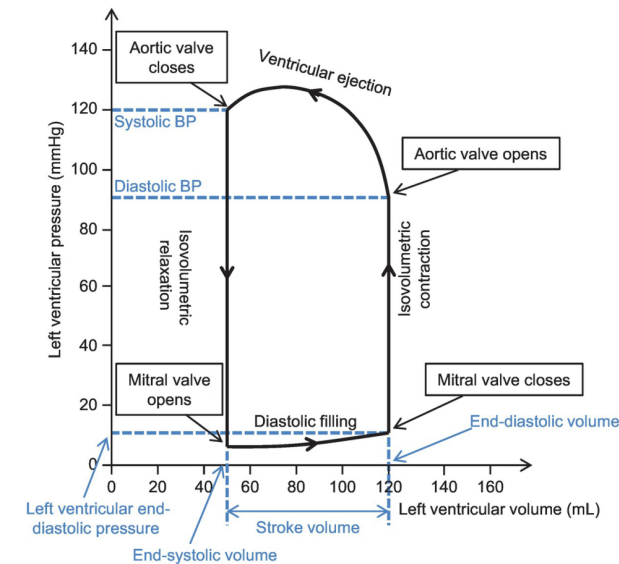
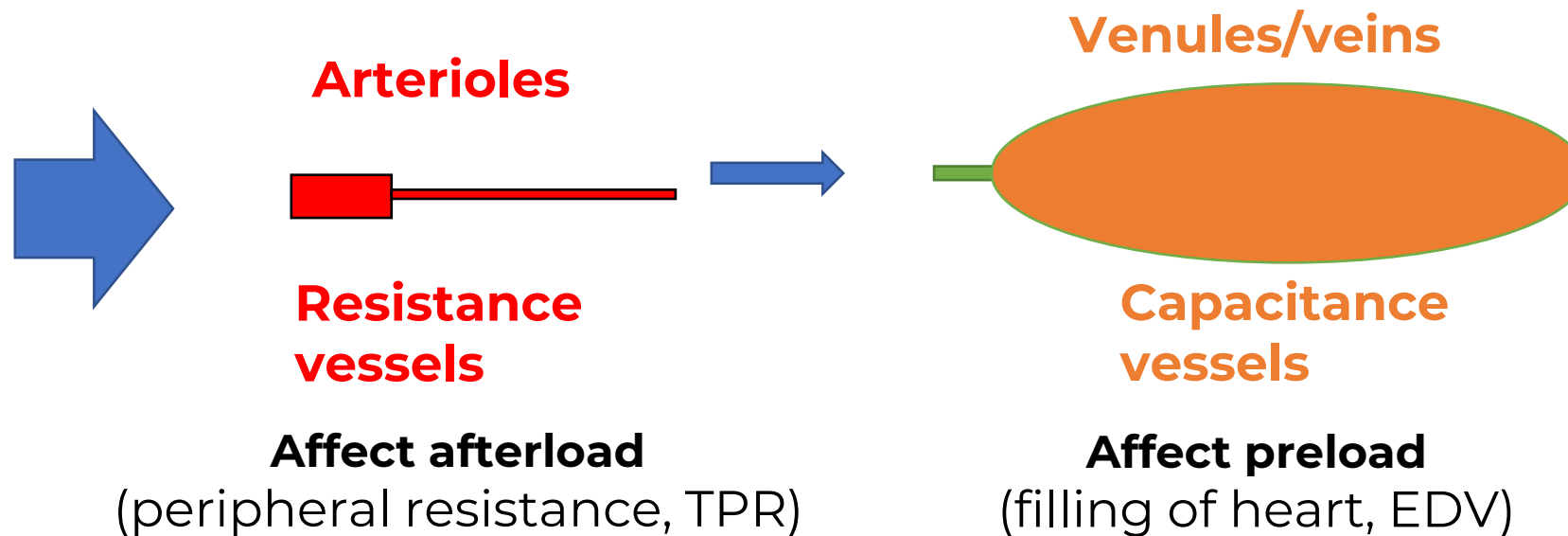
**Short term:** Reduced pressures at capillaries, reduced blood flow to cutaneous tissue in cold stress

**Clinical examples:** Pre-eclampsia → massive vasoconstriction (endothelin). Reduced pressure in capillaries, reduced tissue perfusion.

# Importance of blood vessels in flow and myocardial performance

$$\text{CO} = \frac{\text{MAP} - \text{CVP}}{\text{TPR}}$$

$$\text{MAP} = (\text{CO} \times \text{TPR}) + \text{CVP}$$



**V**asoconstriction/ **V**asodilation – affects afterload

**V**enoconstriction/ **V**enodilation affects preload

## **Explanation of previous slide: How changes in vessel diameter (radius) affects flow.**

Homeostatic mechanisms try and ensure that the pressure head generated by the heart is fairly constant in a resting individual. This ensures that each organ is adequately perfused with blood and oxygen. Darcy's Law states that the flow of blood round the circulation is dependent on this pressure head. But does this mean that the blood flow through each individual organ is fixed and cannot be altered? What if you're in a very cold room and you need to preserve your body heat. One way of doing this is to reduce your skin blood flow. Or if you are exercising, your skeletal muscles need a greater blood flow since they require more O<sub>2</sub> and nutrients. Can such changes in flow be achieved – if the pressure head is relatively constant? Yes, because the resistance to blood flow through individual organs can be altered. Go back to Darcy's Law: blood flow is inversely proportional to resistance, and resistance is related to the diameter of the blood vessel. So should the diameter of skin blood vessels decrease the blood flow through the skin would be reduced, thus conserving body heat. Conversely, if the diameter of skeletal muscle blood vessels increases muscle blood flow would increase. Changes in diameter of blood vessels are brought about by altering the degree of tone in vessel smooth muscles. These smooth muscles are always partially contracted: an increase in contraction results in **vasoconstriction** and a reduced blood flow; whereas a decrease in contraction (relaxation) results in **vasodilation** and a larger blood flow.

### **Points to consider about Resistance**

- Sympathetic NS: innervates and stimulates arterioles causing vasoconstriction.
- Other vasoconstrictors: Endothelin and serotonin
- Vasodilators – Nitric oxide, histamine, adenosine
- Since, Resistance  $\propto 1/r^4$
- Small  $\downarrow$  in arteriolar radius causes a large  $\uparrow$  in resistance.
- Arterioles are called “resistance vessels”

# Lecture Outline

Darcy's Law/ Bulk Flow law (**FLOW =  $\Delta P / R$** )

Poiseuille's Law

- Poiseuille's law which states that flow (Q) is equal to pressure (P) [the change in pressure, or the difference between mean arterial pressure (MAP) and central venous pressure (CVP)] divided by resistance (R)

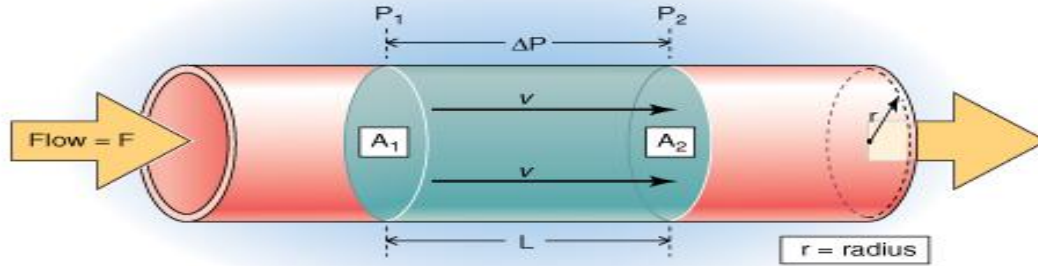
Importance of resistance

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**Velocity, Flow and cross-sectional area**

- **As the blood vessels cross sectional area increases, flow increases, the velocity decreases.**

# Velocity, Flow and Cross-Sectional Area



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A bolus of fluid (blood) moves along a tube with velocity  $V$  with a cross-sectional area  $A$  and a distance of length  $L$

Velocity = displacement of blood / time (cm/second)

$$V = \Delta x / \Delta t;$$

Volume flow = Amount of blood moving a particular distance over time

$$\text{Flow} = \Delta V / t = A \cdot \Delta x / \Delta t$$

$$\text{Flow} = \text{Area} * \text{velocity}$$



## Why Does This Matter?

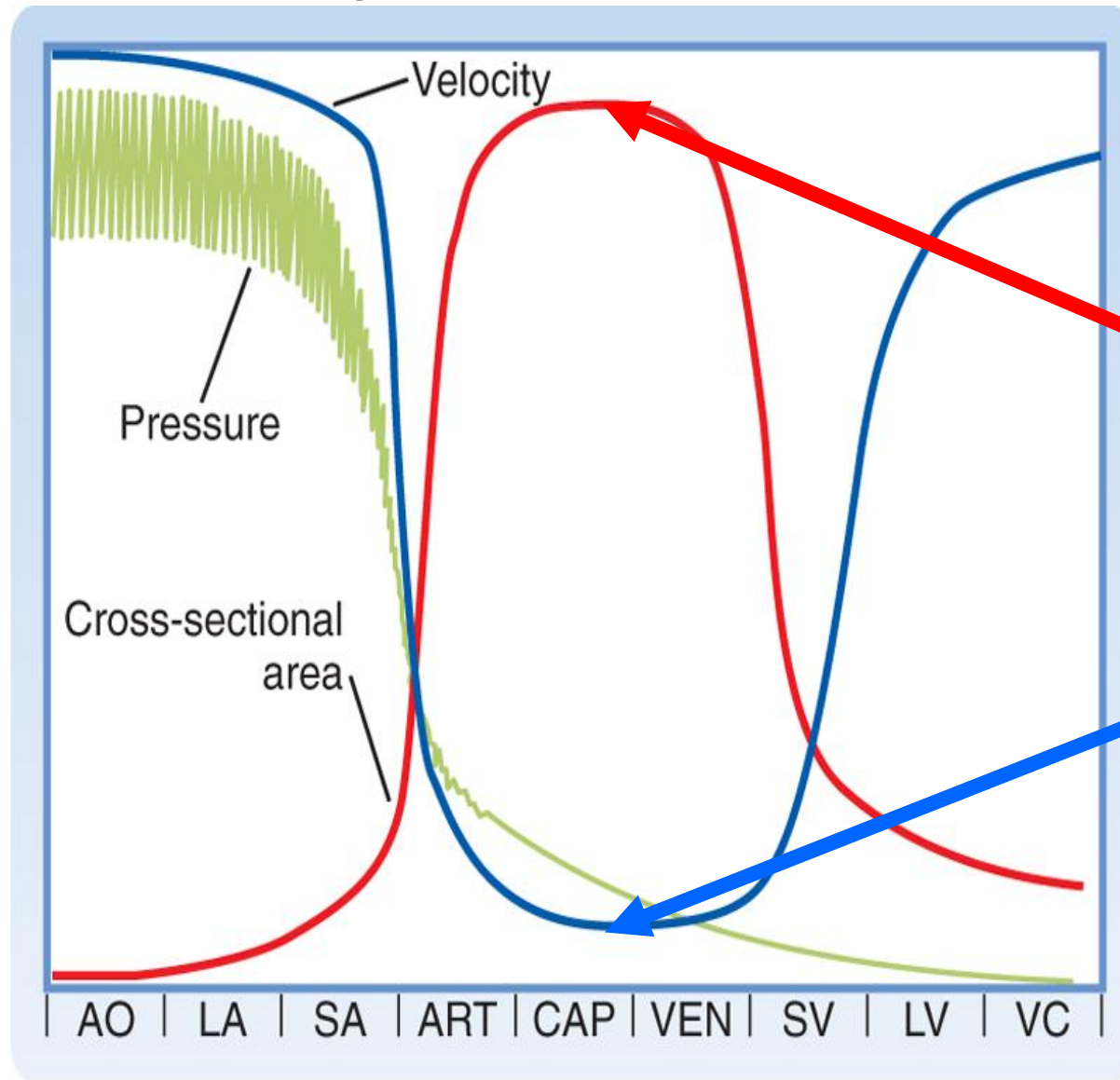
$$\therefore \text{Velocity} = \text{Flow} / \text{Cross sectional Area}$$

**Flow (cardiac output)** stays constant (~5 L/min)

**Area** varies depending on the vessel (e.g., aorta vs capillaries)

So, **velocity changes** as blood moves through vessels of different diameters.

# Velocity of Blood Flow



Velocity (cm/sec) = Flow(cm<sup>3</sup>/sec) / Area  
Same Flow (5L/min) or CO through all vessels

**Capillaries have the largest cross-sectional area**

**Capillaries have the slowest velocity**

Velocity in capillaries is very low vs. aorta; allowing adequate time for diffusion and transfer of nutrients

- Pressures lowest in capillaries- reduce edema

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## **Explanation of the graph on previous slide:**

### **Velocity of Blood Flow and Cross Sectional Area of the Circulation**

The vascular system is made up of arteries, arterioles, capillaries, venules and veins. Let us now consider the total cross sectional area of each component which is the cross sectional area of one vessel times the number of vessels. Which component do you think has the largest cross sectional area? Yes, the capillaries. Although the diameter of each is very tiny, we have millions of them. The aorta has a very large diameter but we only have one! As blood spreads out along the arterial tree the increase in number of vessels due to branching more than outweighs the reduction in vessel diameter. As a result, the total cross sectional area of the circulation increases from the aorta to the capillaries and then decreases as venules converge into veins

### **Effect of cross sectional area on velocity of blood flow.**

Velocity of Blood Flow (cm/s) = Flow (mL/s) / Cross sectional area (cm<sup>2</sup>)

Velocity = Flow (= Cardiac Output) / Area

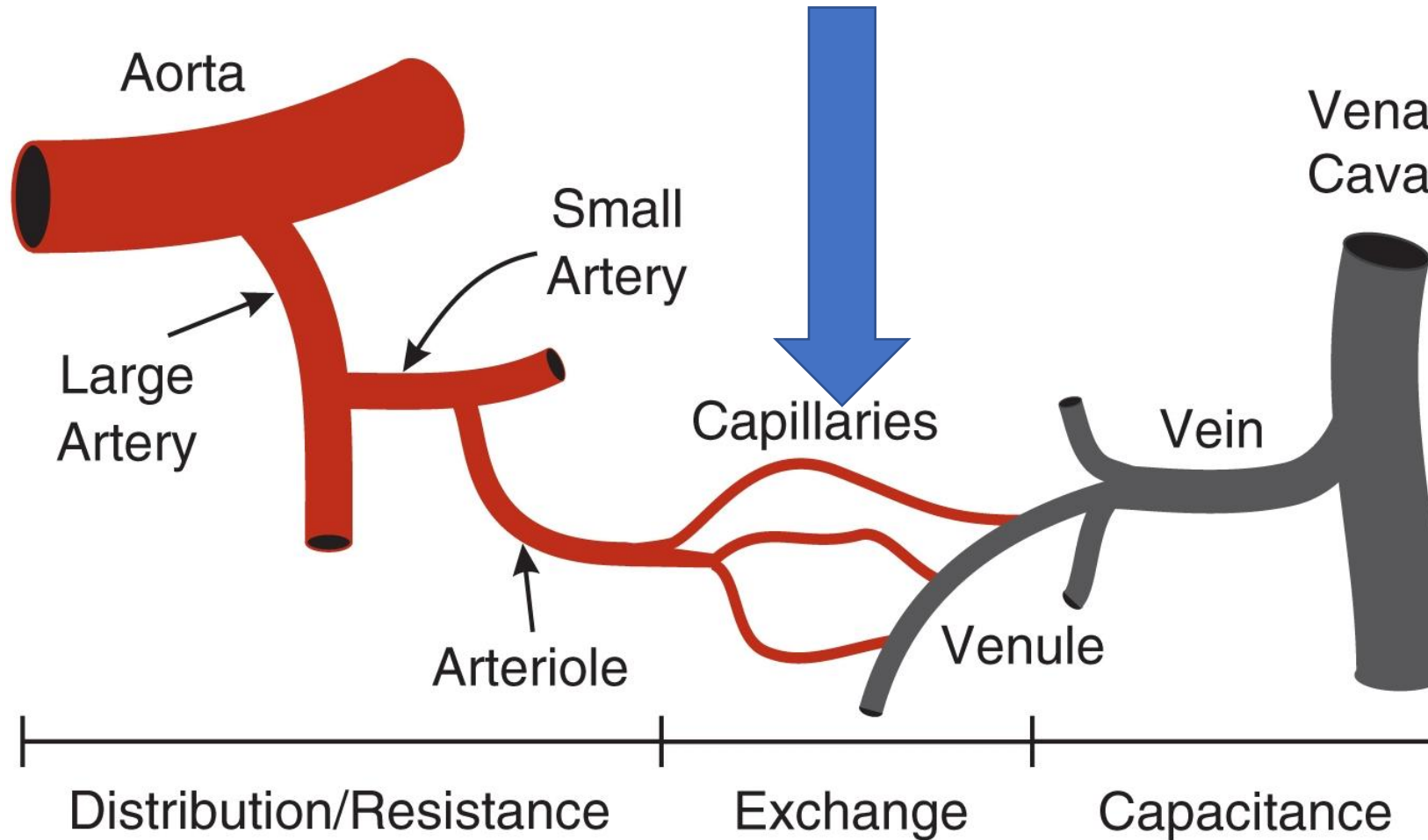
From the above equation, provided that flow is constant, the velocity is inversely related to the cross sectional area. *Which vessels have the largest cross sectional area?* Yes, the capillaries. *Where in the vascular system is the velocity of blood the slowest?* Yes, the capillaries. In fact, the blood travels 200 times more slowly through the capillaries than through the arteries. *Is there an advantage to this?* Yes, the slow velocity through the capillaries allows time for the diffusion of gases and nutrients. The same scenario is found in the pulmonary circulation, where the slow transit time in pulmonary capillaries allows time for the blood to equilibrate with alveolar gases.

Don't confuse flow and velocity. Flow is the volume displaced per unit time. In the dimensional terms mass M, length L and time T: Flow =  $L^3/T$  ; velocity =  $L/T$  but since velocity is flow per unit area =  $L^3/T \times 1/L^2$  which is =  $L/T$ ). Therefore the equation is mathematically correct.

Imagine walking by a river with a constant flow, as the river widens the current (velocity of flow) gets slower, and as it narrows the current is faster.

# SUMMARY IMAGE

Slow velocity  
Parallel (lower resistance)



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# SUMMARY – Key points

- Darcy's law: Blood flow is determined by the pressure differences of cardiac output and vessel compliance or systemic resistance;  $Q = \Delta P / R$
- Poiseuille's law which states that flow (Q) is equal to pressure (P) [the change in pressure, or the difference between mean arterial pressure (MAP) and central venous pressure (CVP)] divided by resistance (R):
- To increase blood flow, either increase the pressure difference (e.g., increased cardiac force) or decrease the systemic vascular resistance (e.g., dilate blood vessels). The size and shape of blood vessels can alter the ease of blood flowing through the blood vessels.
- Blood vessel diameter is a factor for resistance.
- The total resistance for a continuous blood vessel or group of vessels in **series** (one after another) is simply represented by the following, where R1 represents a section of a blood vessel with a given resistance:  $R_t \text{ (total resistance)} = R_1 + R_2 + R_3$
- However, when blood vessels diverge and create multiple paths with a common destination, we say these blood vessels are in **parallel** and resistance takes on the equation  $1/R_t = 1/R_1 + 1/R_2 + 1/R_3$
- In parallel: total resistance is less than the sum of its parts. A physiologic example of this phenomenon can be observed in capillary beds, where although blood vessels are very small, they are arranged in parallel and offer comparatively low resistance to flow.
- E.g. Of clinical importance- management of HTN – vasodilation reduces systemic vascular resistance, decreasing TPR/SVR for a given cardiac output, the pressure is decreased (remembering  $Q = P/R$ , for Q to remain constant as R goes down so must P).